01 / PROBLEM AND CONTRIBUTION

Regular & cascode current mirrors

EEE 433/591 · Analog IC Design · Lab 2 | Fall 2025

RESULT AT A GLANCE

Compared regular and cascode current mirrors, with reported output impedances of 17.73 kΩ and 35.9 kΩ and current ratios near the intended 5:1 scaling.

Engineering question

How does cascoding change current-source output resistance and voltage headroom?

Situation

A current mirror should maintain its programmed current as the output voltage changes. A regular mirror and a cascoded version expose the tradeoff between output resistance, bias complexity, and headroom.

Task

Design and simulate both topologies for a nominal 0.5 mA output from a 0.1 mA reference, then compare DC and small-signal behavior.

My contribution

Sized and biased the mirror circuits, documented DC sweeps and small-signal tests, and compared the two implementations in the submitted lab report.

Tools and concepts

Current mirrors / Cascode biasing / Cadence Virtuoso / Small-signal analysis / AC simulation

02 / DESIGN AND METHOD

From the task to implementation

01 / Implementation

Swept output voltage and identified the reported 0.5 mA crossing: 767.324 mV for the regular mirror and 887.297 mV for the cascode. These crossings are operating-point observations, not measured minimum compliance voltages.

02 / Implementation

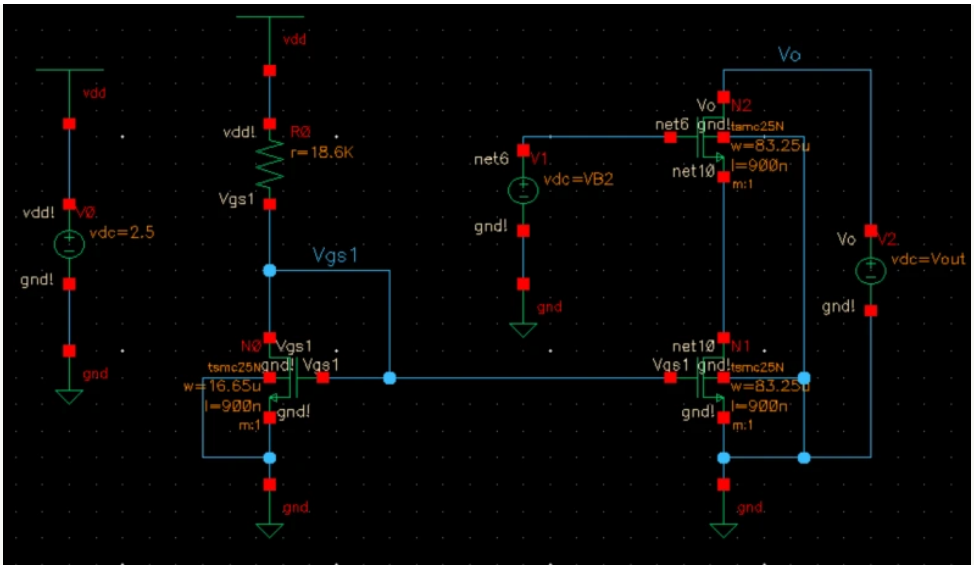
Perturbed the reference branch at 1 kHz and compared reported peak current ratios: 5.062 for the regular mirror and 5.056 for the cascode, against the intended ratio of five.

03 / Implementation

Compared the output-impedance and AC-response results. The final table reports 17.73 kΩ versus 35.9 kΩ and bandwidths of 52.501 kHz versus 51.714 kHz.

Matching the comparison

The mirror ratio concerns reference-to-output scaling; output resistance concerns sensitivity to output voltage. They are different properties and require different testbench perturbations.



Cascode current-mirror schematic

Source report p. 7: cascoded output branch and biasing.

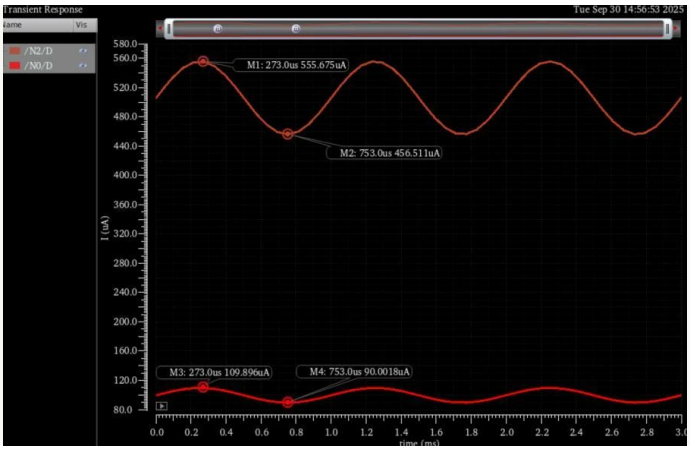
03 / RESULTS AND PERSPECTIVE

Evidence and engineering perspective

Regular / cascode Rout	17.73 / 35.9 kΩ
Reported current ratios	5.062 / 5.056
Output-current target	0.5 mA

Reported result

The final comparison table reports approximately twice the output impedance for the cascode in these simulations, with a slightly lower bandwidth. Both reported peak-current ratios are near five. This is a nominal simulation comparison.



Cascode reference and output currents

Source report p. 11: the 1 kHz current-scaling experiment.

Scope and limitations

The two output-impedance tests use different perturbation amplitudes (10 mV and 20 mV). The report does not establish mismatch, process-corner, or silicon performance. The earlier resume summary is not used for the impedance or bandwidth numbers.

Possible next steps - proposed work

- Repeat both output-impedance tests with the same small-signal amplitude.
- Define a current-error tolerance and extract a compliance-voltage range.

Source material: EEE 433/591 Lab 2 submitted report, physical PDF pp. 3-5, 7, 9-11, 13.